Slide Set #1

Introduction

Pietro Coretto
pcoretto@unisa.it

Econometrics
Master in Economics and Finance (MEF)
Università degli Studi di Napoli "Federico II"

Version: Tuesday 28th December, 2021 (h12:38)

Notes

What is econometrics? XXXX

"*The application of statistical and mathematical methods to the analysis of economic data, with a purpose of giving empirical content to economic theories and verifying them or refuting them*". (Maddala, 1992)

Emphasis on

- Economic theories
- Economic data
- Statistical and mathematical methods

Notes

Typical workflow

- Economic theory (or theories). Example: Keynesian consumption model:
$$C = a + bY$$
- Collect economic data
- Specification of the econometric model:
$$C_i = a + bY_i + \text{random fluctuation}$$
- Inference and/or prediction:
  - testing: $\mathsf{H}_0 : b = 0$ vs $\mathsf{H}_1 : b \neq 0$
  - estimation of the unknowns to do economic analysis: how large is b_{Italy} compared to b_{US}?
  - prediction: if some policy produces $\Delta Y = 10^9$euro, what's the predicted change in C?

Notes

Dependence and causality

Dependence

In terms of probability notions the question is whether $\Pr\{Y \mid X\} \neq \Pr\{Y\}$ or $\Pr\{X \mid Y\} \neq \Pr\{X\}$. Just this. In terms of empirical data we want to know whether variations of Y are somewhat *associated* with variations of X

Causality

Assume that $\Pr\{Y \mid X\} \neq \Pr\{Y\}$, or $\Pr\{X \mid Y\} \neq \Pr\{X\}$, or suppose that data tell us that variations of Y are somewhat associated with variations of X. The question is: why and how this happens? Are we sure that there is a *direct link* between X and Y. Can we exclude that there is an additional Z that is related to both X and Y so that this causes the observed dependence?

Notes

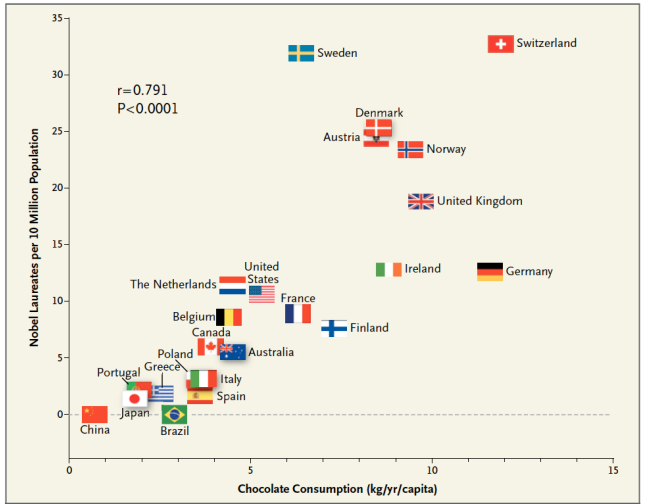


Figure 1. Correlation between Countries' Annual Per Capita Chocolate Consumption and the Number of Nobel Laureates per 10 Million Population.

Source: Messerli, Franz H. 2012. "*Chocolate Consumption, Cognitive Function, and Nobel Laureates*". New England Journal of Medicine. doi:10.1056/NEJMon1211064.

Notes

Predictive modeling

The statistical model "*exploits the dependence*" between an input/feature X and output/outcome variable Y, so that you can predict the outcome value $\hat{Y}_i$ for the ith sample unit for which you only observe the input value X_i.

What's a good model here? Why such prediction works doesn't matter, your problem is to guarantee that the

$$\text{(Squared) Prediction Error} = (\hat{Y}_i - Y_i^{\text{truth}})^2$$

is as small as possible for most sample units.

Predictive modeling is the main paradigm of computer science, artificial intelligence, machine learning, etc.

Notes

Explanatory modeling

The statistical model "*explains and describes*" the link based on which an exogenous/independent variable X determines variations of an endogenous/dependent variable Y.

Usually a theoretical economic model itself provides the causality link.

What's a good model? We look for different requirements here:

- causal parameter(s) of the model, i.e. the parameters linking the X to the Y, uniquely describe the impact of X on Y (indentifiability)
- the model is valid in the sense that the underlying causal hypothesis cannot be rejected based on the empirical data (testing)
- there is a way to use empirical data to estimate the causal parameter(s) of the model uniquely and accurately (bias, consistency, etc.)

Notes

Example: the Keynes' consumption model would be good if

- b "*uniquely*" explains the impact of Y on C
- we can reject $\mathsf{H}_0 : b = 0$ against $\mathsf{H}_1 : b \neq 0$
- there is a way to produce a unique estimate $\hat{b}$ such that

$$\text{Estimation Error} = \hat{b} - b^{\text{truth}}$$

  is small enough in some sense

Explanatory modeling is the main paradigm in econometrics and other social sciences.

In econometrics *Explanatory models* are subdivided in *structural models* and *reduced form models*

Notes

Mixing the paradigms

The two paradigms have different goals, although they share a lot of statistical techniques

⚠ Statistical methods need to be used and interpreted in a different way depending on whether we want to *predict* or *explain*

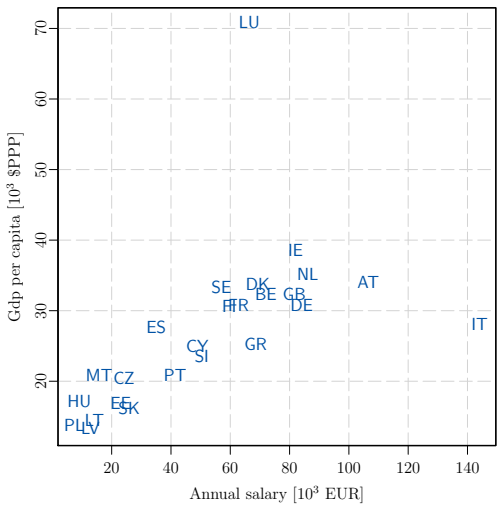
There are fields where traditionally the distinction between these two paradigms is smooth (epidemiology, bio-sciences, etc).

⚠ The modern abundance of massive data collections increased the popularity of the predictive modeling approach. There is a recent tendency to mix the two paradigms unconsciously, but this can easily lead to dramatically wrong scientific statements

Notes

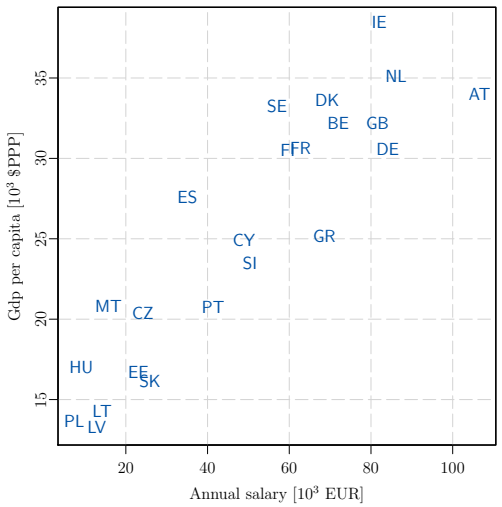
Linear modeling

European parliament salaries



Notes

After cleaning outlying observations



Notes

Joint distribution, covariance and correlation

In previous courses you learned that

- a pair of random variables (X, Y) are linearly dependent (statistical notion) if their joint distribution is such that

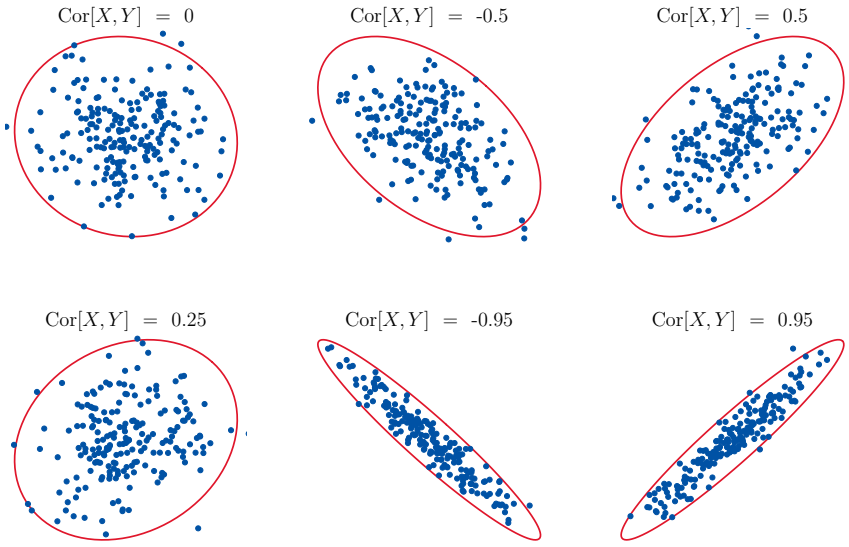
$$\begin{aligned}\mathrm{Cov}(X,Y) &= \mathrm{E}[(X-\mu_X)(Y-\mu_X)] \neq 0 \\ &= \iint (x-\mu_X)(y-\mu_Y)\, f_{X,Y}(x,y)\, dx\, dy \neq 0\end{aligned}$$

- the strength of the dependence is measured by the correlation

$$\mathrm{Cor}[X,Y] = \frac{\mathrm{Cov}[X,Y]}{\sqrt{\mathrm{Var}[X]}\sqrt{\mathrm{Var}[Y]}} \in [-1,1]$$

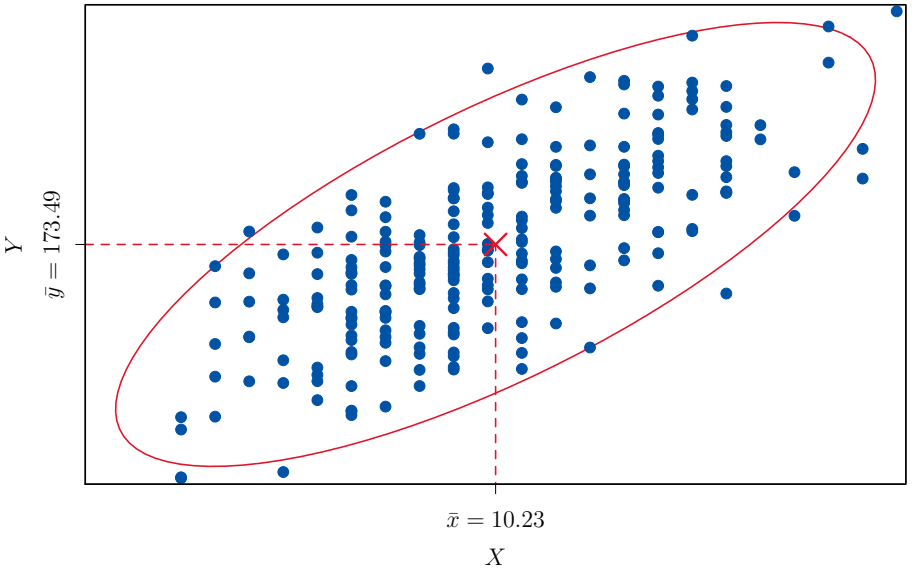
- independence $\implies \mathrm{Cor}[X,Y] = 0$, the converse is not true
- $\mathrm{Cor}[X,Y] = \mathrm{Cor}[Y,X]$ the symmetry of the covariance operator doesn't allow to make causal statements

Notes



source

Notes



Notes

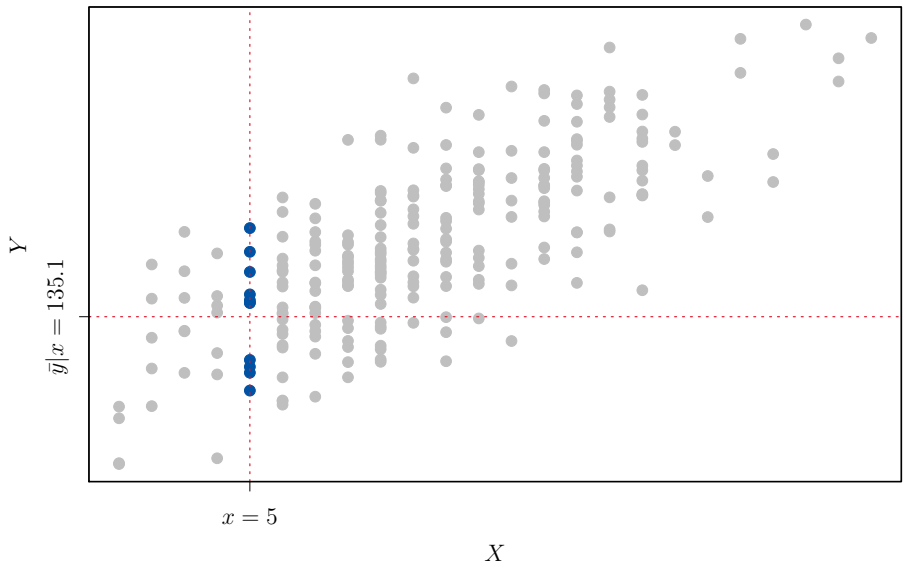
If (X, Y) are correlated, a sample from their joint distribution most of the times produces a scatter where the majority of the points lie in an ellipsoidal region centered at sample means $(\bar{x}, \bar{y})$

The volume of the ellipse captures the overall joint dispersion (multivariate variance)

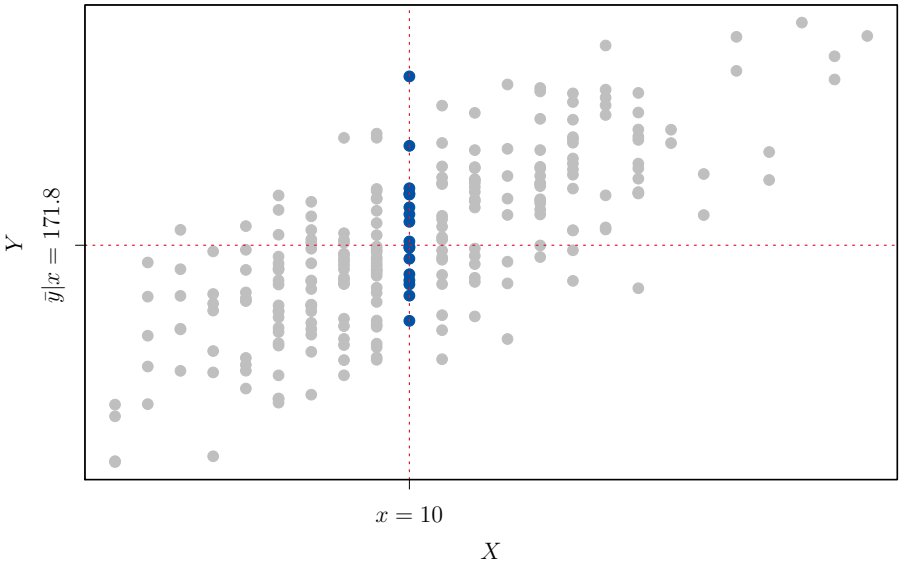
Highly correlated pairs ($\mathrm{Cor}[X, Y]$ close to ± 1) have scatters compressed along the main axes of the ellipses

How do we model this? A statistical model for (X, Y) needs to be able to reproduce this behavior of the joint distribution. What kind of sampling design can reproduce such a thing?

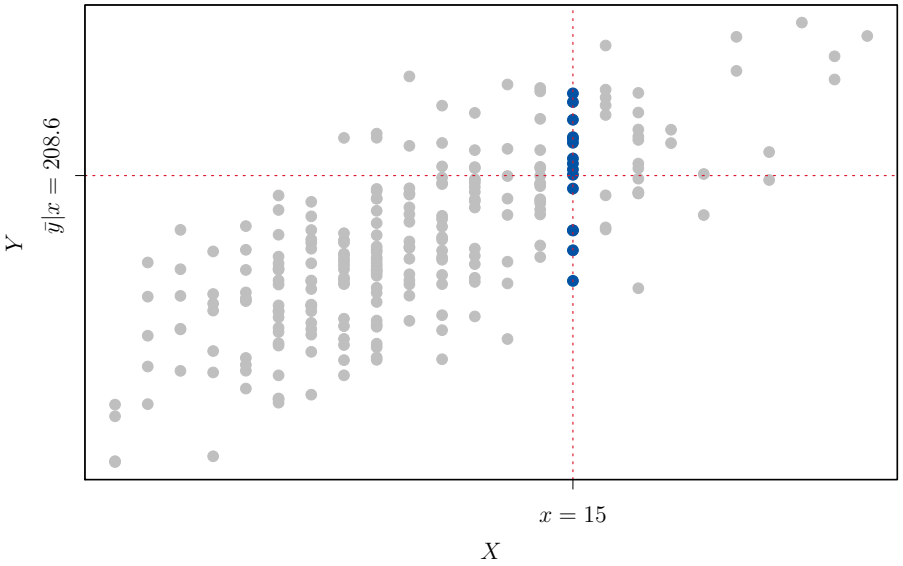
Notes



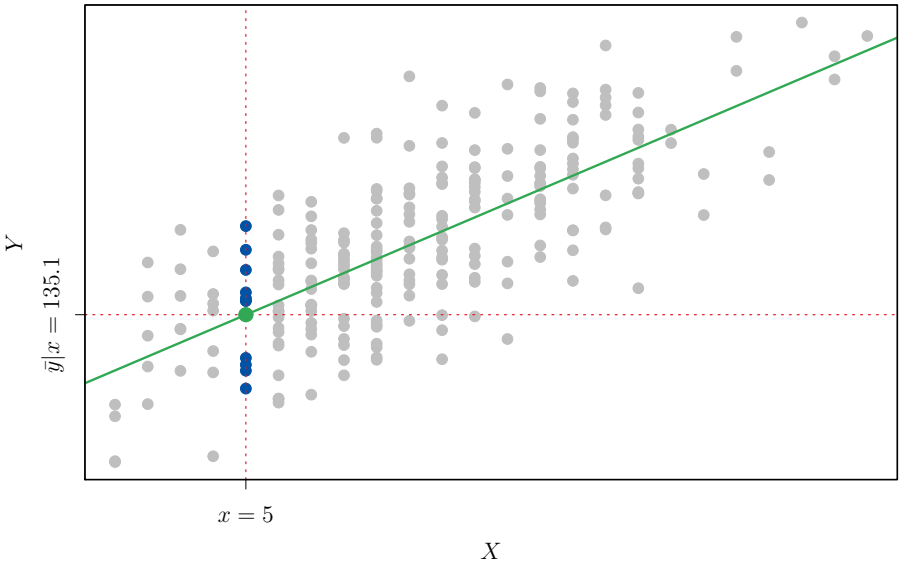
Notes



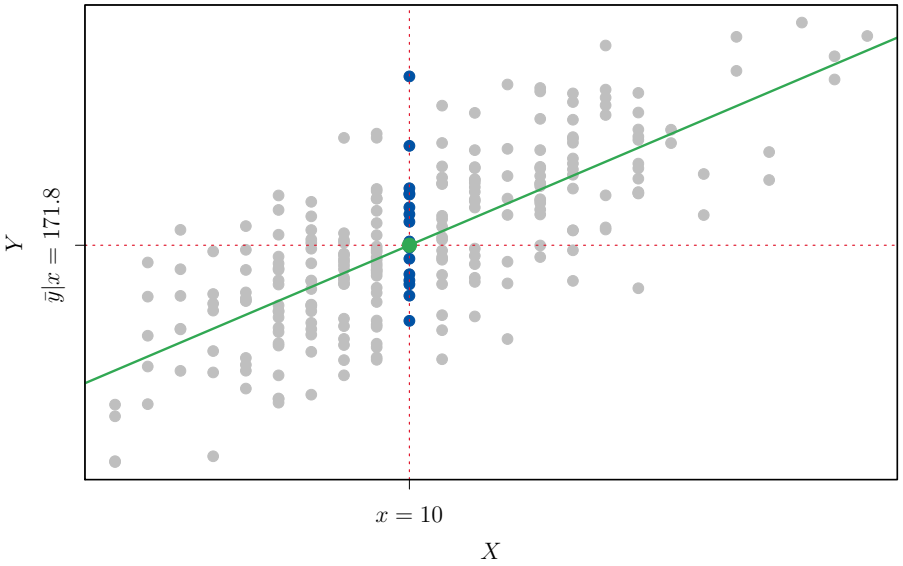
Notes



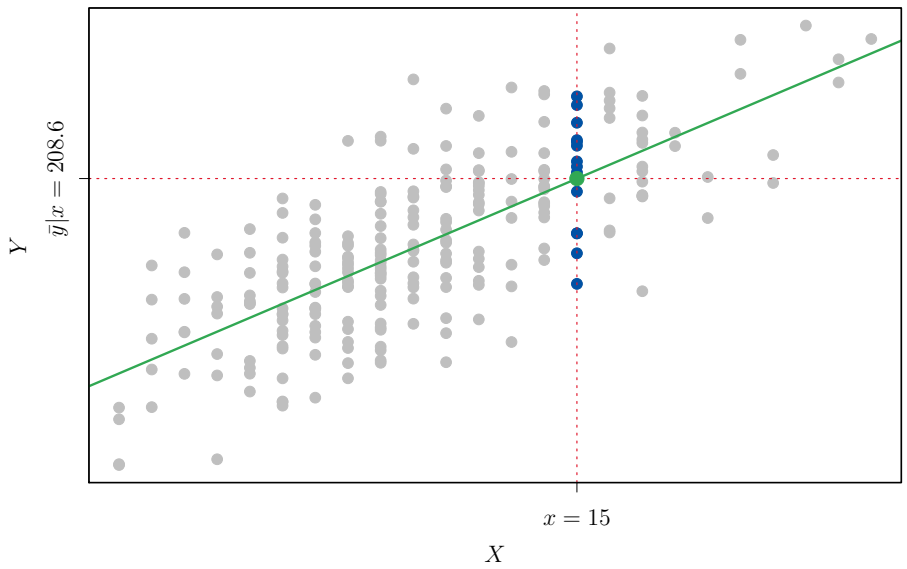
Notes



Notes



Notes



Notes

The conditional mean of $Y \mid X$ *increases proportionally* as we increase X

For a fixed $X = x$, the points are randomly scattered around the conditional mean of $Y \mid X = x$

Therefore if (Y_i, X_i) are pairs of random variables sampled from a joint distribution for which $\mathrm{Cor}[X, Y] \neq 0$, a model to represent what we would observe from such distribution would be

$$\begin{aligned} Y_i &= \mathrm{E}[Y|X_i] + \text{random fluctuation} \\ &= \beta_0 + \beta_1 X_i + \text{random fluctuation} \end{aligned}$$

The latter is a linear regression model. It will be the object of interest of this course. It allows to reproduce the observed correlation. This can be used to *predict* or to *explain*

Notes

Correlation twists in modern big-data

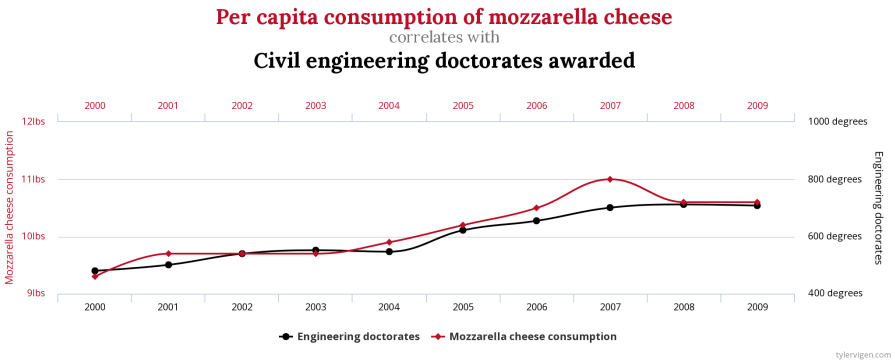
Big data = massive collection of data with huge dimensions in both

- n =number of sample units
- p =number of variables/features measured on each unit

Issues relevant for econometric applications

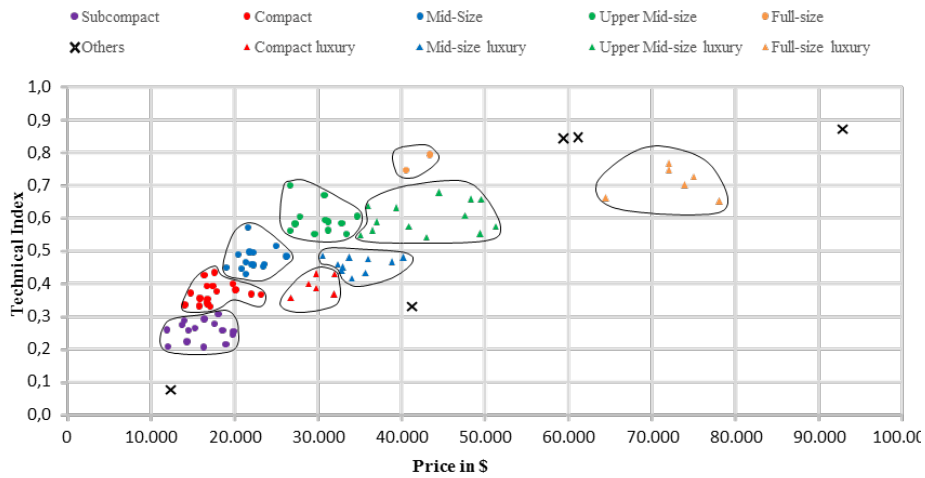
- n much smaller than p
- spurious dependence
- heterogeneity

Notes



source: `http://tylervigen.com/spurious-correlation`

Notes



source: `https://www.allaboutlean.com/automotive-market-strategy/`

Notes

Collecting data

We are flooded with data

"Good" econometric analysis crucially depends on the availability of "good" economic data. Key ingredients are: a) good data sources; b) appropriate data processing.

In modern days the ability to perform sophisticated data (pre)processing tasks is essential. Computing is a necessary tool for modern econometrics.

Major issues

- Relevant economic variables are not always observable (e.g. expectations)
- Wide gap between the ideal measure and its observable counterpart (e.g. stock of capital in production functions)
- Data acquisition frequency is often not appropriate
- Non-response can be serious if respondents are not a random sample drawn from the population ("selectivity bias" problem)

Notes

Types of data vs indexing of the units

Suppose $X =$`GDP`

- Cross-section: X is measured on the ith unit at a given time point, i is a country, an individual, an household, etc.
  Example: $i = 1 =$`UK`, $i = 2 =$`Italy`, X_1 GDP of UK in 2018, $X_2 =$GDP of Italy in 2018
- Time series: X is measured on a certain unit at a time point i, i can be a year, month, quarter, etc.
  Example: $X_{1960} =$ GDP of Italy in 1960, $X_{1961} =$ GDP of Italy in 1961, etc.
- Panel or longitudinal data: i, j are time and unit indexes. X is measured on the ith unit in the jth period.
  Example: $X_{1,1980}$ is the GDP of UK in 1980, $X_{2,2000}$ is the GDP of Italy measured in 2000.

Notes

Types of variables vs levels/labels

Level/label: value recorded for each sample unit.
Example: $X_i = 100$, 100 is the level, or $G_i =$`Male`, `Male` is the label/level

- Quantitative (aka numerical) variables: levels are numerical, differences between levels have a metric interpretation. Numerical variables can be: continuous (e.g. income), discrete (e.g. number of patents), and discretized continuous variables (e.g. age)
- Ordinal variables: levels can be ordered, differences between levels don't have a metric interpretation.
  Example: $Y =$`income`$=\{$ `upper`$=3$, `middle`$=2$, `lower`$=1\}$, therefore `lower`$\preceq$`upper`, but *upper-lower* = 3-1 = 2 = ?
- Categorical (aka nominal) variables: labels cannot be ordered, differences between labels don't have a metric interpretation.
  Example: $S=$`sex`$=\{0=$`Male`, $1=$`Female`$\}$, can we say `female`$\preceq$`male` or `male`$\preceq$`female`? Moreover, *Female -Male* = 1-0 = 1 = ?

Notes